

03c. Eclipse: Setup & IDE Configuration



Page is about?

Purpose: Guide new SDETs on how to install and configure Eclipse for automation test development using the PRIME framework (Java, Maven, Selenium, TestNG, Cucumber).

Covers project import, encoding, ANT setup, plugins, and Perforce integration.

Pre-Requisites

Tool	Setup Steps
Java 17+	- Download and install from Oracle/OpenJDK - Set JAVA_HOME system variable C:\Program Files\Java\jdk-17.0.2 - Add %JAVA_HOME%\bin to system Path - Ensure no other conflicting Java references exist - Verify via Command Prompt: set or java -version
Maven	- Install from: https://maven.apache.org/install.html - Set MAVEN_HOME and add %MAVEN_HOME%\bin to system Path - Verify via: mvn -v

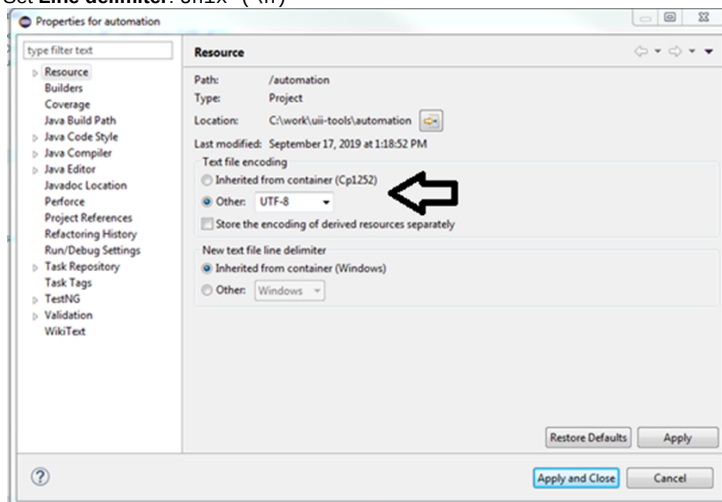
1. Eclipse Installation

Task	Details
Download Eclipse IDE	Eclipse Download – Eclipse IDE for Java Developers
Architecture Match	Ensure your Eclipse version matches Java bit version (32-bit/64-bit)
Install Location	Suggested: C:\Eclipse

2. Import Automation Project

1. Open Eclipse
2. Go to: File Open Projects From File System Select Import Source Directory
3. Browse to: <Choose any local repository folder> Select automation (Git Repo) folder
4. Complete import
5. Right-click project Properties Resource

- Set **Text file encoding**: UTF-8
- Set **Line delimiter**: Unix (\n)



3. Add External JARs (Recommended)

Step	Action
Locate JARs	L:\Departments\Financial Services\UI\UI\UI-QA\Cucumber_Automation\automation-env Simply copy the newest version for each of these to your computer (C:\ recommended): • ant • cucumber • selenium - this also includes a "startSeleniumGrid.bat" file which needs to be copied • testng • groovy From Local: automation-env folder, include these jars: prime\lib jars, selenium, cucumber, testng, groovy, tools.jar (from JDK)
Add to Path	Right-click Project Build Path Configure Build Path Add External JARs

4. Configure ANT for Compile/Run

Step	Action
Add View	Window Show View Other... ANT
Add Build Files	From each project (astro, unite, jett):

- Add: <project>/build.xml (compile)
- Add: <project>/bin/build.xml (run) || Repeat | Map 6 total build.xmls if working across all projects |

5. Switch to Navigator View (Optional)

Step	Action
Enable View	Window Show View Navigator

6. Perforce Integration (No longer in Use)

Step	Action
Install Plugin	Location: L:\Departments\Financial Services\UI\UI\UI-QA\Cucumber_Automation\automation-env\perforce
Automation-env folder:	
Share Project	Right-click project Team Share Project Perforce
Server	herring:1818
Auth	Use your Ascensus credentials + selected workspace

7. Cucumber Plugin (For Feature Navigation)

Step	Action
Install	Eclipse Marketplace Search Cucumber Eclipse Plugin
Configure	Right-click project Configure Convert to Cucumber Project
Use	Navigate easily between .feature and step definitions

Tip: The more Java files your project has, the heavier this plugin can become. Pull only what you need from Perforce/GitLab.

8. Post-Import Reminders

Action	Description
Restart Eclipse	To reflect environment variables (e.g., JAVA_HOME, ANT_HOME)
Verify Encoding	UTF-8 Properties > Resource
Check Java Compiler Compatibility	Project > Properties > Java Compiler > Compiler compliance level = 17
Run Sample Test Suite	Use <code>testng.xml</code> or <code>mvn clean install</code> to validate setup

9. Folder Mapping – ANT vs Maven

Tool	Folder	Notes
ANT	<code>build.xml</code> / <code>bin/build.xml</code>	Compile and run automation manually
Maven	<code>pom.xml</code>	Use for dependency management, TestNG execution, CI/CD